

step 1

Collect real-world research papers and train a reward model.

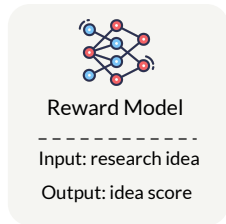
Collect peer-reviewed papers from online platforms.



Extract ideas or proposals from **real-world** research papers and their review scores.



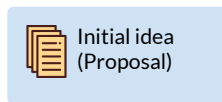
Train language models with the research idea and review score.



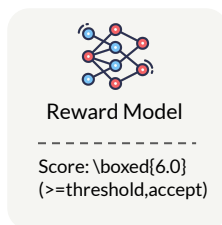
step 2

Launch AlphaResearch with original ideas and programs

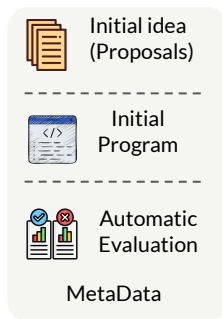
A proposal to describe the research problem.



Evaluate proposal as positive idea or negative ones using reward models.



Construct initial program and automatic evaluation (metadata) for the idea.



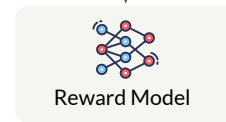
step 3 (Iteration 1~n)

Iterate the autonomous research process

Generate new idea based on the MetaData (proposals, program, evaluation results) at **Round n-1**



Evaluate proposal as positive idea or negative ones using reward models.



Update the program to match the new idea. And evaluate it with execution.



Obtain new MetaData and enter the next iteration

